

ADITYA CHUNDURI

(678) 776-0805 | chunduri@usc.edu | linkedin.com/in/aditya-chunduri | github.com/Chunduri-Aditya
chunduri-aditya.github.io/Portfolio

Education

University of Southern California

Master of Science in Applied Data Science

Los Angeles, CA

January 2024 – December 2025

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering (AI/ML Specialization)

Chennai, India

June 2019 – July 2023

Technical Skills

Languages: Python, C++, SQL, JavaScript, Bash

ML/AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Keras, OpenCV, Librosa, Demucs

MLOps and Cloud: Docker, AWS, FastAPI, Flask, Git, GitHub Actions, CI/CD, Gradio, LangChain

Data and Tools: PostgreSQL, ChromaDB, Neo4j, Pandas, NumPy, Plotly, Ollama

Experience

USC – Viterbi School of Engineering

Los Angeles, CA

Research Assistant, Computer Vision and Medical Imaging

August 2024 – December 2024

- Engineered a U-Net segmentation pipeline for retinal artery-vein classification on the CVD-Masks dataset, achieving 94% Dice score through architectural tuning and custom loss functions.
- Designed modular PyTorch training pipelines with experiment tracking across 3 concurrent projects, reducing iteration cycles by 30% and enabling reproducible model comparisons.
- Built an automated medical image scraper with quality-validation controls, cutting manual curation time by 40% across 10,000+ images.

SSN College of Engineering

Remote

Research Intern, Computer Vision and Object Detection

June 2021 – July 2021

- Built real-time object tracking pipelines achieving 30+ fps; fine-tuned YOLO-based CNN architectures improving detection robustness by 15% across 5 environmental conditions.
- Produced 10+ reusable PyTorch and TensorFlow modules standardizing deployment workflows and eliminating 100% of manual frame-annotation overhead.

Projects

Agent Shield: LLM Agent Evaluation Framework | Python, Inspect AI, AgentDojo

April 2026 – Present

- Designing a reproducible evaluation framework for LLM agents with a unified metric schema across 8 attack surfaces; built on UK AISI's Inspect AI harness.
- Benchmarking across 8 frontier models (Claude, GPT-4.1, Gemini 2.5, Llama 3.1, Mistral Large, DeepSeek V3) with reproducible seeds, judge model versioning, and CI-ready eval logs.

AI RemixMate: AI-Powered DJ Mixing Engine | Python, PyTorch, Librosa, Demucs, Gradio

September – December 2025

- Engineered an end-to-end audio ML pipeline: Demucs source separation, Librosa spectral feature extraction, and Camelot-Wheel harmonic matching; deployed via FastAPI with GPU acceleration.
- Shipped a Gradio web UI supporting 8+ audio formats with real-time BPM, key, and energy analysis; open-sourced with 9 GitHub stars, confirmed via landscape analysis as the only pipeline integrating all components.

AkashicTree: Automated Content Generation Pipeline | Python, Diffusers, Ollama, ElevenLabs, FLUX.1

- Designed a model-agnostic ACG (Automated Content Generation) pipeline coordinating text (Ollama), image (FLUX.1/Diffusers), and audio (ElevenLabs) models from a single input brief, cutting production time by 80%.
- Implemented tiered backend-switching between local Ollama inference and cloud APIs, reducing per-iteration inference costs by 90% while maintaining output quality.

AI Health Journal: Privacy-First LLM Assistant | Python, Flask, Ollama, ChromaDB

January – May 2025

- Deployed a 7B-parameter LLM inference pipeline entirely on-device via Ollama; optimized ChromaDB vector retrieval for sub-second context injection across 10,000+ document chunks.
- Applied DPO (Direct Preference Optimization) fine-tuning to align model outputs with user health context with zero cloud dependency or data egress.

Model-Behavior-Lab: LLM Benchmarking Platform | Python, Ollama, Plotly, CI/CD

September – December 2024

- Built an LLM evaluation platform executing 5,000+ automated behavioral tests across 10+ models (GPT-4, Claude, Llama, Mistral) with CI/CD integration for continuous benchmarking on model version updates.
- Designed modular JSON test suites enabling plug-and-play evaluation of any new model, cutting evaluation setup from hours to under 5 minutes per release cycle.

Publication

Wind Power Analysis Using Machine Learning – IJRASET, August 2023 (Co-Author): Co-developed a neural network framework integrating numerical weather data for wind energy forecasting across multiple datasets.